Safety text indicators are excluded from all decision rules (no validated labels)

Empirical records

Linked records
flight logs, removals, work orders
1,711 tracked unscheduled removals

Frequency and consequences
frequency λ_a (removals per FH)
work per event m_a (complete cases)
parts value per event c_a
missing work kept missing

Retrospective evaluation
13 rolling origins, top-3 selection frequency vs $P_a(w)$, $w \in [0, 1]$
coverage of observed work and parts value; work envelopes

Totals repeat frequency
total work $= \lambda_a m_a$
total parts value $= \lambda_a c_a$
zero-stock EBO $\propto \lambda_a$ (same lead time)
normalisation cannot remove this

Assumed logistics

Assumed inputs
removal rates, QPA, standard values, 35-day replenishment

pipeline = demand rate × lead time
outputs not added to the priority

Corrected METRIC
$\mathrm{EBO}(s) = \mathrm{E}[(X - s)^+]$; fill $\Pr(X < s)$
budget and service scenarios
integer benchmark
Poisson vs NB demand

Synthetic policy

Synthetic system
8 assets, Weibull lives, purchase lead time, explicit downtime cost

Joint age–stock selection
replacement age T, base stock s
150 policies, pilot seeds
candidates frozen

Independent evaluation
50 new paired seeds
12 one-factor scenarios
adapted and transferred policies
cost and availability